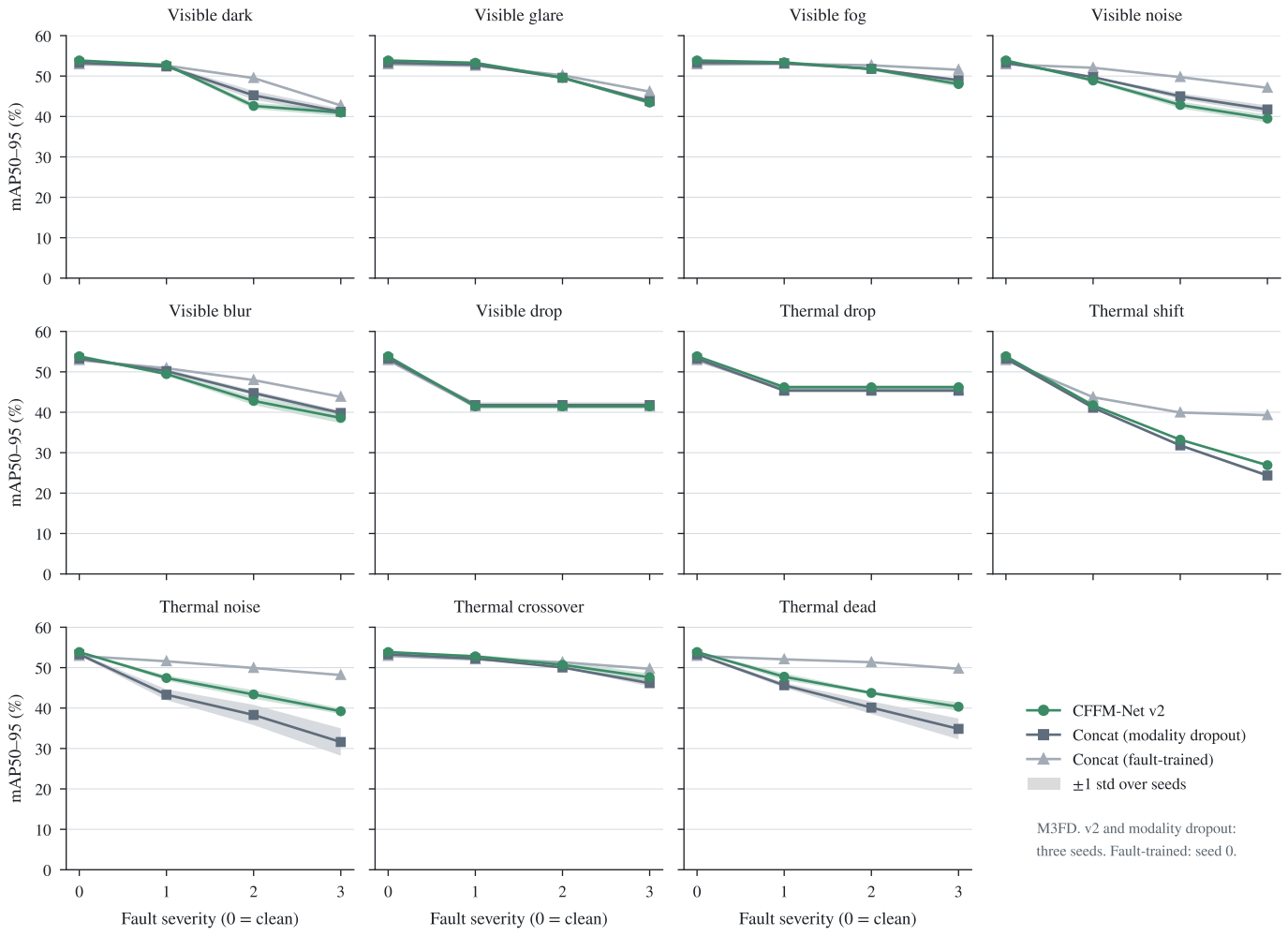
Visible dark
Visible glare
Visible fog
Visible noise
Visible blur
Visible drop
Thermal drop
Thermal shift
Thermal noise
Thermal crossover
Thermal dead
mAP50–95 (%)
0
10
20
30
40
50
60
0
1
2
3
Fault severity (0 = clean)
CFFM-Net v2
Concat (modality dropout)
Concat (fault-trained)
±1 std over seeds
M3FD. v2 and modality dropout:
three seeds. Fault-trained: seed 0.